Inverse energy cascade in self-gravitating collisionless dark matter flow and effects of halo shape

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ABSTRACT

In dark matter flow, haloes grow by absorbing single particles and smaller haloes, so mass is passed from small to large haloes: an inverse mass cascade. We ask how energy is passed along with the mass, and whether the change of halo shape plays the role that vortex stretching plays in turbulence. We define the flux of any specific quantity V through the halo mass scale m_h and show that its value at the smallest scale is $\varepsilon_V = -(\mu + \lambda_V - s)M_h H\langle V\rangle$, where $M_h \propto a^\mu$ is the mass in haloes, $\langle V\rangle \propto a^{\lambda_V}$ its mass-weighted mean and s the growth exponent of V in haloes of fixed mass. We measure all terms in an Einstein–de Sitter simulation with 256^3 particles. (i) Kinetic energy, both of the motion of haloes and of the motion inside haloes, is carried to larger haloes (inverse cascade); the inter- and intra-halo potential energies are carried to smaller haloes (direct cascade). The fluxes are independent of m_h, for haloes of up to a few hundred particles, only at $z \leqslant 0.3$. (ii) With the measured exponents the formula reproduces every flux to within 2 per cent at $z < 2$; the exponents of the simplest steady-state model fail for the inter-halo potential and the radial kinetic energy. (iii) The system has not reached the steady state of the cascade picture: M_h grows as $a^{1/2}$ only near $z = 0$ (local exponent 0.47), and the total kinetic and potential energies grow as $a^{1.13}$ and $a^{1.19}$ at $z \approx 0$, not as $a^{3/2}$. (iv) Haloes are elongated, $r_{\lambda 3}/r_{\lambda 1} \approx 2$–3, but their moment of inertia exceeds that of a sphere of equal volume only by a median factor 1.60; their spin axis lies within 30° of the plane perpendicular to the major axis for 70 per cent of haloes with $n_p \geqslant 20$ (isotropic: 50 per cent). Shape change therefore cannot drive the energy cascade; mass accretion does.

Key words: dark matter – large-scale structure of Universe – methods: numerical – methods: analytical

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1 INTRODUCTION

In the cold dark matter model, structure forms hierarchically: small haloes form first and grow by accreting the surrounding matter and by merging into larger haloes [1, 2, 3, 4]. The statistics of this growth, the merger rates and the mass accretion histories of haloes, are among the best-studied results of N-body simulations [5, 6, 7, 8]. They matter for observations because the assembly history of a halo shapes both its internal structure and the galaxy it hosts: the accretion rate sets the concentration of the halo [5, 9, 10], and the gas that falls in with the dark matter feeds the growth of the central galaxy [11, 12]. Accretion also sets the energy of haloes. The velocity dispersion of the dark matter follows a tight virial scaling with halo mass, which underlies the dynamical mass estimates of galaxy clusters and their use as a cosmological probe [13], and the departure from virial equilibrium, measured by the ratio of kinetic to potential energy, is the usual criterion for selecting relaxed haloes in simulations [9, 14].

Haloes are not spherical. Simulations find them triaxial and mostly prolate, with axis ratios that depend on mass and on the accretion history; massive haloes, which formed recently, are the least spherical [15, 16, 17, 18, 19]. Halo shapes are measured directly in the weak-lensing shear maps of galaxy clusters [20] and, for the Milky Way, from stellar streams [21]. The angular momentum of haloes, acquired through tidal torques [22, 23], sets the size of the galactic discs that form inside them [24], and its direction is correlated with the halo shape and with the surrounding cosmic web [25, 26]. These correlations are the physical origin of the intrinsic alignments of galaxies, one of the main systematic effects in cosmic-shear surveys [27, 28].

These studies describe how the mass, shape and spin of haloes evolve, but not how energy is passed between haloes of different mass as they grow. In turbulence, this flux of energy across scales is the central quantity [29, 30, 31]. In three dimensions, kinetic energy passes from large to small eddies (a direct cascade), and the mechanism is the stretching of vortices along their spin axis: a stretched

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vortex spins faster, so its rotational energy rises while its angular momentum is conserved [32, 33]. In two dimensions, where vortices cannot be stretched, kinetic energy passes to large scales instead [34]. Dark matter flow differs from a fluid: it is collisionless, compressible and self-gravitating [35]. Its elementary structures are haloes [36, 37], and because haloes grow by merging, mass is passed from small to large haloes in an inverse mass cascade, which accounts for the halo mass function and the density profiles of haloes [38, 39, 40, 41]. Two questions follow, and they bear directly on the energies, shapes and spins discussed above. How fast is energy passed between haloes of different mass, and in which direction? And does the change of halo shape play the part of vortex stretching, that is, does it drive the energy transfer?

In this paper we answer both questions with an Einstein–de Sitter simulation of 256^3 particles. We define the flux of any specific quantity through the halo mass scale and show that its value at the smallest scale is set by the mass flux and three measurable growth exponents (Section 3). Measuring them for six specific energies, we find that the kinetic energy, both of the motion of haloes and of the motion inside them, is carried to larger haloes, while the inter- and intra-halo potential energies are carried to smaller haloes; with the measured exponents the flux formula reproduces every flux to within 2 per cent at $z < 2$ (Sections 4 and 5). The simplest model of the cascade, a statistically steady state in which the halo mass grows as $a^{1/2}$ and the energies as $a^{3/2}$, is not reached by $z = 0$: the local growth exponent of the halo mass is 0.47, and the total kinetic and potential energies grow as $a^{1.13}$ and $a^{1.19}$. The coherent radial and rotational motion inside haloes also cascades to larger haloes but carries little energy (Section 6). From the shapes of 1,674,583 haloes at $z = 0$ we find that haloes are elongated, yet their moment of inertia exceeds that of a sphere of equal volume only by a median factor 1.60, and their spin axis lies preferentially perpendicular to the major axis (Section 7). Shape change is therefore not the mechanism of the energy cascade; mass accretion is.

The paper is organized as follows. Section 2 describes the simulation and the halo groups. Section 3 defines the flux of a specific quantity through the mass scale and derives its value at the smallest scale. Section 4 describes the energies of the halo and out-of-halo subsystems, and Section 5 measures the energy fluxes and tests the flux formula. Section 6 treats the radial and rotational motion inside haloes and Section 7 the halo shape and spin orientation. Section 8 ranks the evidence and lists further tests, and Section 9 concludes.

2 SIMULATION AND HALO GROUPS

We use the Virgo consortium SCDM run [42, 43]: an Einstein–de Sitter (EdS) universe with the parameters of Table 1. In EdS, $a = (t/t_0)^{2/3}$ and $H = H_0 a^{-3/2} = 2/(3t)$. Haloes are found with a friends-of-friends algorithm (comoving linking length $0.2\,\mathrm{Mpc}/h$, i.e. $b = 0.214$ of the mean interparticle separation) and contain at least two particles; particles in no halo form the out-of-halo subsystem. Halo statistics are computed for halo groups: all haloes with the same number of particles n_p at one redshift. They are public as CSV tables for ten redshifts [44, 45, 46], and every number of this paper is computed from those tables and, for halo shapes, from the per-halo catalogue at $z = 0$ (1,674,583 haloes).

For each halo, the velocity u_h is the mean peculiar velocity of its particles, and the internal dispersion σ^2_{vh} is the one-dimensional variance of the particle velocities about u_h. For a halo group, σ^2_v is the mean of σ^2_{vh} and σ^2_h is the one-dimensional variance of u_h over the haloes of the group. The kinetic energy of a particle therefore has two parts: that of the motion of its halo (σ^2_h) and that of the motion inside its halo (σ^2_v); for the particles of one group $\sigma^2 = \sigma^2_h + \sigma^2_v$. In the same way the specific potential energy of a particle, $\phi = \frac{1}{2}\sum_j V(r_{ij})/m_p$, has an intra-halo part ϕ_v (pairs in the same halo) and an inter-halo part $\phi_h = \phi - \phi_v$. All energies are physical and per unit mass, in $\mathrm{km^2\,s^{-2}}$. The potential is measured relative to that of the nearly uniform $z = 10$ snapshot. Summed over the three subsystems below, the one-dimensional dispersions reproduce the rms peculiar velocity $u^2(z)$ of the simulation to within 1.2 per cent.

Table 1. Parameters of the simulation.

Run	Ω_0	Λ	h	Γ	σ_8	L (Mpc/h)	N	m_p ($M_\odot/h$)	l_{soft} (kpc/h)
SCDM1	1.0	0.0	0.5	0.5	0.51	239.5	256^3	2.27×10^{11}	36

3 THE FLUX OF A SPECIFIC QUANTITY THROUGH THE MASS SCALE

Mass flux. Let $m_g(m_h,t)$ be the mass of the halo group of halo mass $m_h = n_p m_p$ and $M_h(t) = \sum m_g$ the mass in haloes. The mass flux through m_h is the rate at which mass leaves all groups below m_h for the groups above it:

$$\Pi_m(m_h,t) = -\frac{\partial}{\partial t}\int_{m_h}^{\infty}\frac{m_g(m,t)}{m_p}\,\mathrm{d}m. \tag{1}$$

Its derivative, the transfer function $T_m = \partial\Pi_m/\partial m_h = \partial m_g/(m_p\partial t)$, is the rate of change of the group mass. Therefore

$$\Pi_m(m_h,t) = \varepsilon_m + \int_0^{m_h} T_m\,\mathrm{d}m = -\int_{m_h}^{\infty} T_m\,\mathrm{d}m,$$
$$\varepsilon_m \equiv \Pi_m(0,t) = -\frac{\mathrm{d}M_h}{\mathrm{d}t}. \tag{2}$$

In words: mass enters the halo subsystem at the smallest scale at the rate $|\varepsilon_m|$; where the group masses do not change ($T_m = 0$, the propagation range) the flux is independent of m_h; where they grow ($T_m > 0$, the deposition range) the flux falls to zero. A negative flux means mass moves to larger haloes: the inverse mass cascade. If $M_h \propto a^\mu$, then $\varepsilon_m = -\mu M_h H$. Note that ε_m must be kept in equation (2): without it the flux would vanish in the propagation range.

Flux of a specific quantity. Haloes carry with them specific quantities V (per unit mass): kinetic energies, potentials, the energy of rotation. Some of the change of $\sum_{m>m_h} m_g V$ is not transfer at all but the change of V inside haloes that stay above m_h. Removing that part leaves the flux of V through m_h:

$$\Pi_V(m_h,t) = \int_{m_h}^{\infty}\frac{m_g}{m_p}\frac{\partial V}{\partial t}\mathrm{d}m - \frac{\partial}{\partial t}\int_{m_h}^{\infty}\frac{m_g}{m_p}V\,\mathrm{d}m = -\int_{m_h}^{\infty} T_m V\,\mathrm{d}m. \tag{3}$$

The quantity is carried by the mass: its transfer function is $T_m V$. At the smallest scale,

$$\varepsilon_V \equiv \Pi_V(0,t) = M_h\langle\frac{\partial V}{\partial t}\rangle - \frac{\mathrm{d}\,(M_h\langle V\rangle)}{\mathrm{d}t}, \tag{4}$$

where $\langle\cdot\rangle$ is the mass-weighted mean over all haloes. Three exponents describe the growth: $M_h \propto a^\mu$, $\langle V\rangle \propto a^{\lambda_V}$, and s, defined by $\langle\partial V/\partial t\rangle = sH\langle V\rangle$ (for $V = a^s F(m_h)$, s is the growth exponent of V in haloes of fixed mass). Then

$$\varepsilon_V = -(\mu + \lambda_V - s)M_h H\langle V\rangle, \qquad \frac{\varepsilon_V}{\varepsilon_m\langle V\rangle} = \frac{\mu+\lambda_V - s}{\mu}. \tag{5}$$

Equation (5) is exact for power laws; with local exponents it holds at every time. The flux of V equals $\varepsilon_m\langle V\rangle$ only if $\lambda_V = s$, that is

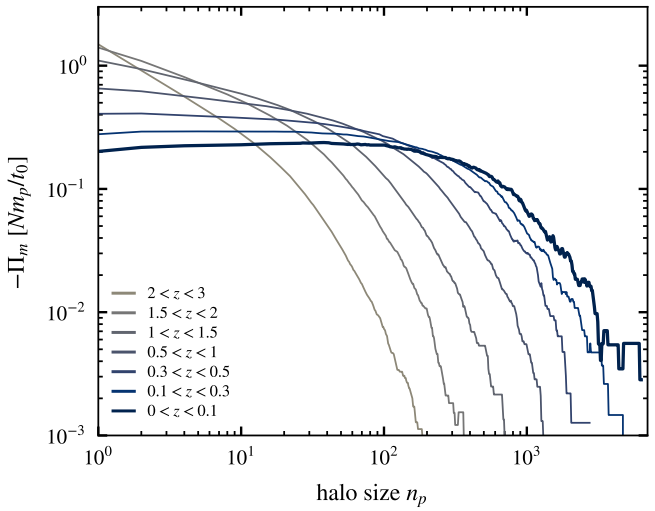


Figure 1. Mass flux $-\Pi_m$ through the halo size n_p between consecutive snapshots (equation 6 with $V = 1$). The flux is negative (inverse cascade); a range of n_p with constant flux (propagation range) forms after $z \approx 0.5$ and widens with time.

if V does not depend on halo mass. *The steady-state model.* In the simplest picture of the cascade, a statistically steady state [38], the halo mass grows as $M_h \propto a^{1/2}$ ($\mu = \frac{1}{2}$), every mass-weighted energy grows as a ($\lambda_V = 1$), and s follows from how V scales with time at fixed mass: $s = 1$ for the motion of haloes and the inter-halo potential ($\sigma_h^2, \phi_h \propto a$), $s = -1$ for the motion inside haloes and the intra-halo potential ($\sigma_v^2, \phi_v \propto a^{-1} m_h^{2/3}$), $s = -3$ for the kinetic energy of radial motion inside haloes and $s = 0$ for that of rotation. The model then predicts $\varepsilon_V/(\varepsilon_m \langle V \rangle) = 1$ for σ_h^2 and ϕ_h and 5 for σ_v^2. Section 5 measures all three exponents and tests these predictions.

Typical haloes. With $M_h = m_h^L n_h^L n_p^L/(1-\lambda)$ for a typical halo of mass $m_h^L \propto a^{\mu/(1-\lambda)}$ [38], equation (5) reads $\varepsilon_V = -(\mu + \lambda_V - s) H m_h^L N_h^L V^L$ with $N_h^L = M_h \langle V \rangle / (m_h^L V^L)$: the flux of any quantity is set by the mass accretion of typical haloes, the counterpart of vortex stretching in this flow.

From two snapshots. The data give m_g and the group means of V at snapshots 1 and 2. The discrete form of equation (3) is

$$\Pi_V(n_p) = -\sum_{n>n_p} \frac{m_{g2}(n) - m_{g1}(n)}{m_p\,\Delta t} \frac{V_1(n) + V_2(n)}{2}, \qquad (6)$$

with V of the other snapshot used for a group that is empty at one of the two. Fluxes are given in units of $Nm_p u_0^2/t_0$ (Nm_p/t_0 for mass), with $u_0 = 354.61\,\mathrm{km\,s^{-1}}$, $t_0 = 13.04\,\mathrm{Gyr}$ and $\Delta t = t_2 - t_1$.

4 ENERGIES OF THE TWO SUBSYSTEMS

Fig. 2 shows the group quantities. Small haloes carry their kinetic energy in their motion ($\sigma_h^2 \gg \sigma_v^2$), large haloes inside them; the cross-over is at $n_p \approx 100$ ($z = 0$). σ_h^2 depends little on n_p up to a few hundred particles. Its mass-weighted mean is $\langle \sigma_h^2 \rangle = \beta_{\sigma h} a u_0^2$ with $\beta_{\sigma h} = 0.79$, 0.70 and 0.57 at $z = 3$, 1 and 0, so $\langle \sigma_h^2 \rangle \propto a^{0.71}$ ($z \leqslant 1$) rather than $\propto a$. The internal dispersion grows with mass as $\sigma_v^2 \propto n_p^{0.70}$ ($100 \leqslant n_p \leqslant 1000$, $z = 0$) and, at fixed n_p, decreases with time as $a^{-0.71}$; $\sigma_v^2 \approx \beta_{\sigma v} a^{-1} u_0^2 n_p^{2/3}$ with $\beta_{\sigma v}$ between 0.024 and 0.030 ($z \leqslant 2$). The intra-halo potential follows the internal dispersion, $-\phi_v \propto n_p^{0.66}$, and the inter-halo potential $-\phi_h$ is nearly independent of n_p for small haloes.

Mass. The halo subsystem holds 6 per cent of the mass at $z = 3$ and 61 per cent at $z = 0$ (Fig. 3a). Its growth slows: the local exponent $\mathrm{d}\ln M_h/\mathrm{d}\ln a$ falls from 3.44 ($2 < z < 3$) to 0.47 ($0 < z < 0.1$). The value $\mu = \frac{1}{2}$ of the steady-state model is therefore reached only at $z \approx 0$; over $z \leqslant 1$ a single power law gives $M_h \propto a^{0.79}$.

Virial equilibrium. The virial ratio $\gamma = -3\sigma^2/\phi$ is one when the virial theorem $2K + W = 0$ holds. Inside haloes, $\langle \gamma_v \rangle$ rises from 1.11 to 1.27 between $z = 3$ and 0 (Fig. 3b): internal equilibrium is set up early, and the excess over one comes from mass accretion and the surface pressure of haloes [39]. For the motion of haloes, $\langle \gamma_h \rangle$ rises from 0.54 to 1.10: equilibrium between haloes is reached only at $z \approx 0.5$. The out-of-halo particles have $\langle \gamma_o \rangle$ = 1.8–2.5, i.e. their kinetic and potential energy nearly cancel. Their specific energy $\frac{3}{2}\sigma_o^2 + \phi_o$ is $20264\,\mathrm{km^2\,s^{-2}}$ at $z = 0$, 17 per cent of their kinetic energy, and it changes sign between $z = 2$ and 1.5; it is small but not conserved.

Energy. The kinetic energy of the halo subsystem overtakes that of the out-of-halo particles at $a \approx 0.5$ (Fig. 4), the potential energy already at $a \approx 0.36$. The out-of-halo kinetic and potential energies per unit system mass change little after $z = 2$ (exponents 0.17 and 0.31 at $z \approx 0$). The total kinetic energy grows with a local exponent that falls from 1.32 to 1.13 between $z \approx 2$ and 0, the total potential energy from 1.41 to 1.19. Neither approaches $\frac{3}{2}$, the value of the steady-state model ($K \propto t$); the same result was found for the whole system from the cosmic energy equation [47]. The energy grows more slowly than t because the halo mass fraction saturates while the specific energies grow more slowly than a.

5 THE ENERGY CASCADE

Mass. Fig. 1 shows the mass flux. It is negative at every n_p and every time. At early times it falls off at once: there is no propagation range. For $0.1 < z < 0.3$ it stays within 10 per cent of ε_m for $2 \leqslant n_p \leqslant 100$ and falls to half at $n_p = 410$; for $0 < z < 0.1$ the half-value point is at $n_p = 716$. The measured ε_m agrees with $-(\mathrm{d}\ln M_h/\mathrm{d}\ln a) M_h H$ to within 1 per cent (Table 2).

Kinetic and potential energy. The four energy fluxes (Fig. 5) have fixed signs over the propagation range: the kinetic energies of the motion of haloes and of the motion inside them are carried to larger haloes ($\Pi_{kh}, \Pi_{kv} < 0$, inverse cascade), and the inter- and intra-halo potentials to smaller haloes ($\Pi_{\phi h}, \Pi_{\phi v} > 0$, direct cascade). As a halo grows its kinetic energy rises and its potential energy falls, so mass accretion moves kinetic energy up and potential energy down the mass scale; the two are tied by the virial ratios of Section 4. The internal energies have the widest propagation range: for $0 < z < 0.1$, Π_{kv} and $\Pi_{\phi v}$ stay within 1 per cent of their value at $n_p = 1$ up to $n_p = 100$. The flux of σ_h^2 is the least uniform: for $0 < z < 0.1$ it is 1.3 times larger at $2 \leqslant n_p \leqslant 20$ than at $n_p = 1$, because small groups still lose mass at that time, and it changes sign in the deposition range at $z < 0.3$ and $z > 1.5$. The intra-halo potential carries the largest flux; at $z \approx 0$ the internal kinetic energy flux is 13 times that of the motion of haloes (Table 2).

Test of the flux formula. For each specific energy and each pair of snapshots we measure μ and λ_V from the mass-weighted totals and s from the first term of equation (4). With these exponents equation (5) reproduces the measured ε_V of all six energies to within 2 per cent for the six intervals at $z < 2$ and to within 8 per cent for $2 < z < 3$, where M_h grows by a factor of 2.7 within one interval; the difference is the error of the two-point exponents. The content of the formula is therefore in the exponents (Fig. 6):

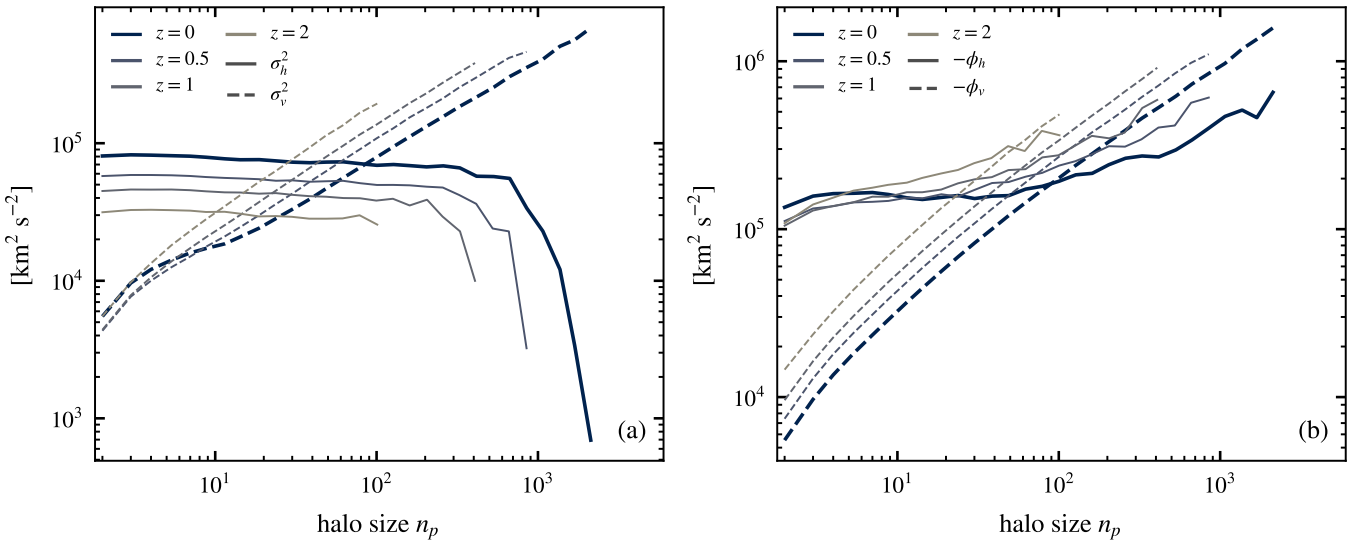


Figure 2. Halo groups of n_p particles at $z = 0$, 0.5, 1 and 2 (halo-weighted means in logarithmic bins). (a) Velocity dispersion of the haloes, σ_h^2 (solid), and mean internal dispersion, σ_v^2 (dashed). (b) Inter-halo potential $-\phi_h$ (solid) and intra-halo potential $-\phi_v$ (dashed). The motion of haloes carries the energy of small haloes, the motion inside haloes that of large haloes.

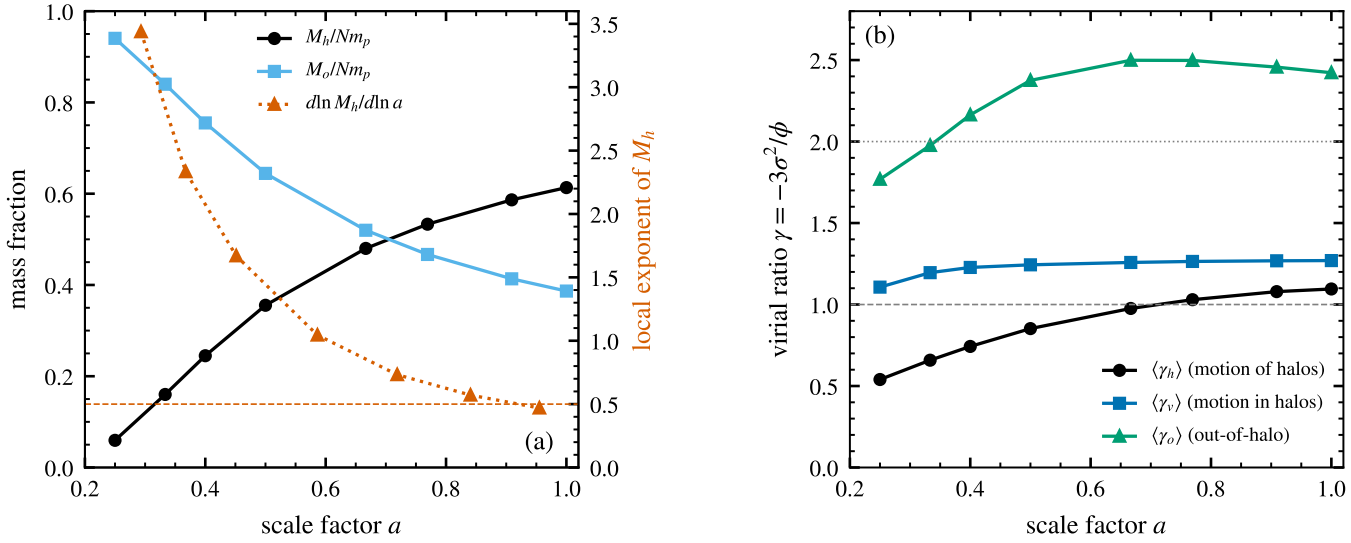


Figure 3. (a) Mass fractions of the halo (M_h) and out-of-halo (M_o) subsystems and the local growth exponent $\mathrm{d}\ln M_h/\mathrm{d}\ln a$ (triangles, right axis; dashed line $\frac{1}{2}$). (b) Mass-weighted virial ratios $\gamma = -3\sigma^2/\phi$ of the motion of haloes (γ_h), the motion inside haloes (γ_v) and the out-of-halo particles (γ_o); $\gamma = 1$ is virial equilibrium, $\gamma = 2$ zero total energy.

Table 2. Fluxes at the smallest scale, $\varepsilon = \Pi(n_p = 1)$, between consecutive snapshots, in units of Nm_p/t_0 (mass) and $Nm_pu_0^2/t_0$ (energies), and the growth exponent of the halo mass. Subscripts: *kh*, *kv* kinetic energy of the motion of and inside haloes; ϕh, ϕv inter- and intra-halo potential; *urp*, *ua* kinetic energy of the radial and rotational motion inside haloes.

Interval	a_{mid}	$\mathrm{d}\ln M_h/\mathrm{d}\ln a$	ε_m	ε_{kh}	ε_{kv}	$\varepsilon_{\phi h}$	$\varepsilon_{\phi v}$	ε_{urp}	ε_{ua}
$2 < z < 3$	0.293	3.44	−1.490	−0.334	−0.294	1.932	0.752	−0.080	−0.041
$1.5 < z < 2$	0.367	2.33	−1.402	−0.373	−0.449	2.016	1.111	−0.074	−0.035
$1 < z < 1.5$	0.451	1.67	−1.099	−0.338	−0.547	1.760	1.335	−0.065	−0.027
$0.5 < z < 1$	0.586	1.04	−0.653	−0.231	−0.603	1.245	1.454	−0.053	−0.018
$0.3 < z < 0.5$	0.719	0.73	−0.407	−0.149	−0.623	0.940	1.494	−0.053	−0.014
$0.1 < z < 0.3$	0.841	0.57	−0.278	−0.097	−0.625	0.734	1.492	−0.038	−0.012
$0 < z < 0.1$	0.955	0.47	−0.202	−0.049	−0.608	0.601	1.455	−0.026	−0.011

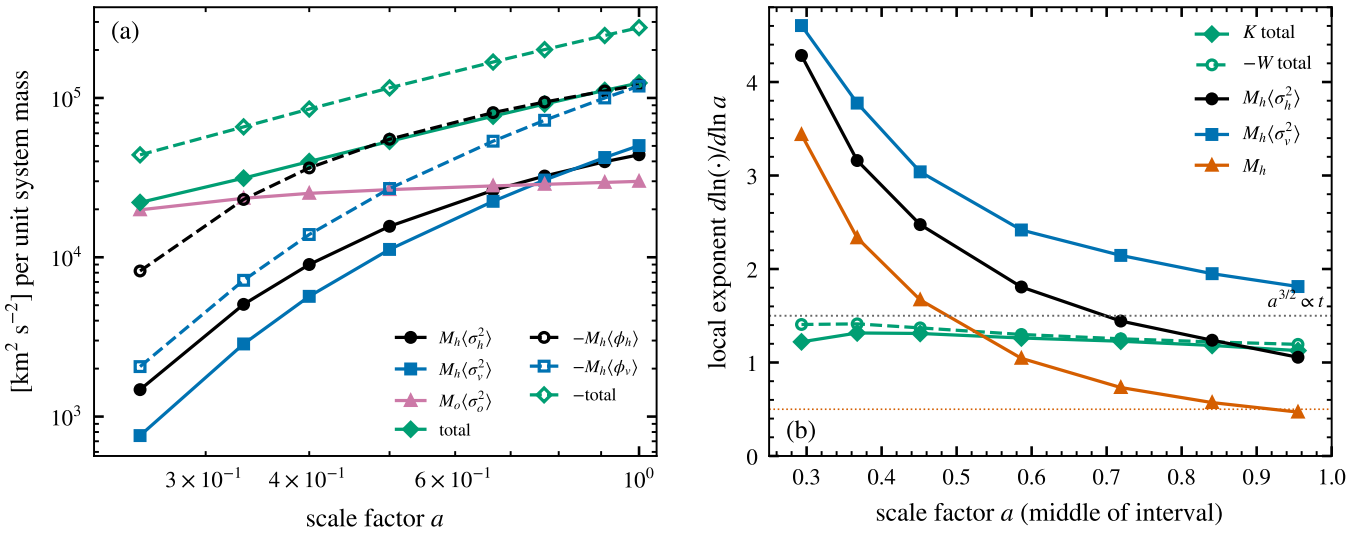


Figure 4. (a) One-dimensional dispersions (solid) and potentials (dashed, sign reversed) per unit system mass: motion of haloes $M_h\langle\sigma_h^2\rangle$, motion inside haloes $M_h\langle\sigma_v^2\rangle$, out-of-halo particles $M_o\langle\sigma_o^2\rangle$, and totals ($M$ in units of Nm_p). (b) Local growth exponents. The halo subsystem dominates after $a\approx 0.5$; the totals grow as $a^{1.1\text{–}1.4}$, more slowly than $a^{3/2}$ ($\propto t$).

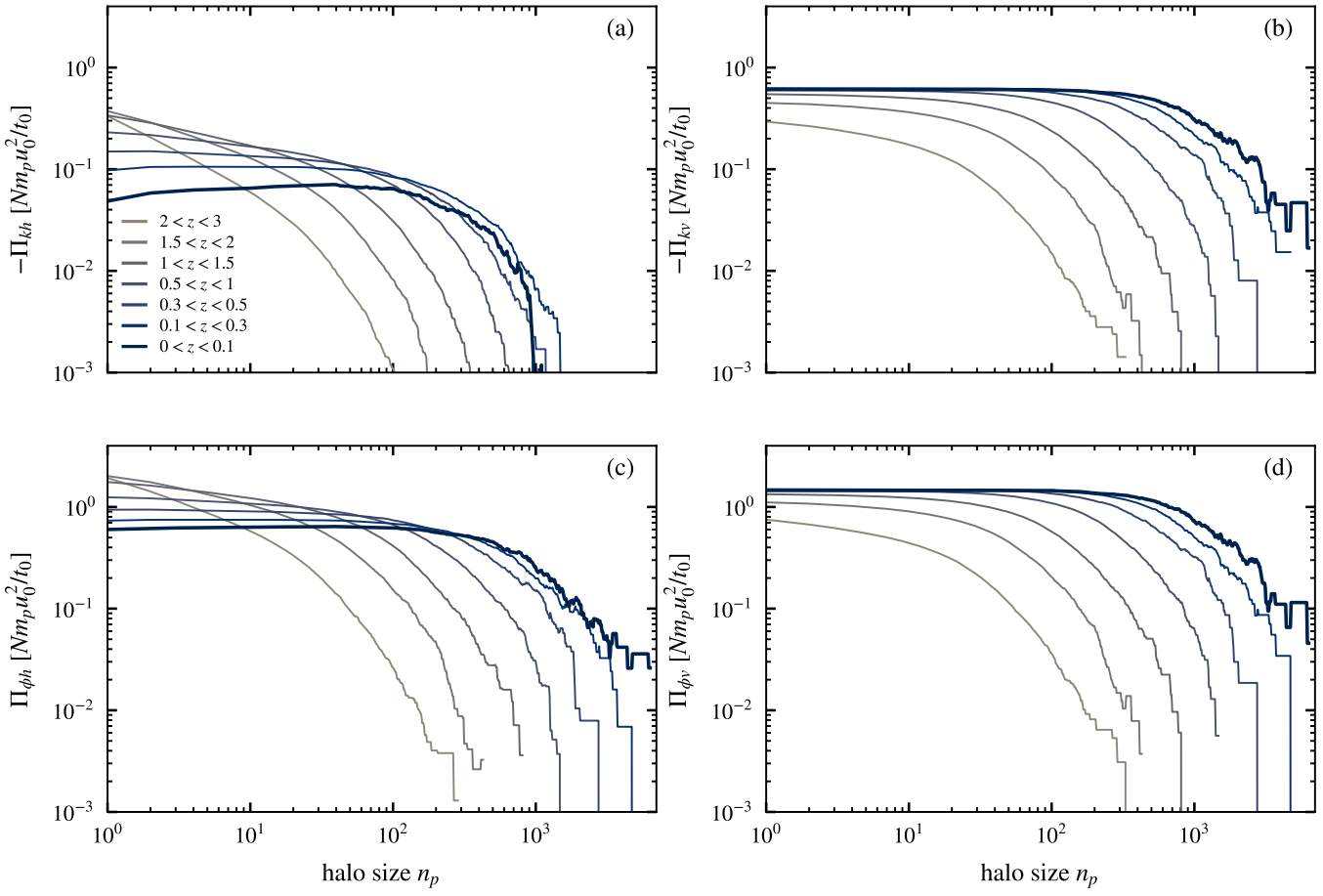


Figure 5. Flux functions of the specific energies through the halo size n_p, from consecutive snapshots: (a) kinetic energy of the motion of haloes, $-\Pi_{kh}$; (b) kinetic energy of the motion inside haloes, $-\Pi_{kv}$; (c) inter-halo potential, $\Pi_{\phi h}$; (d) intra-halo potential, $\Pi_{\phi v}$. Kinetic energy is carried to large haloes, potential energy to small haloes. The fluxes are independent of n_p over a range of n_p only at late times, most clearly for the internal energies (b, d).

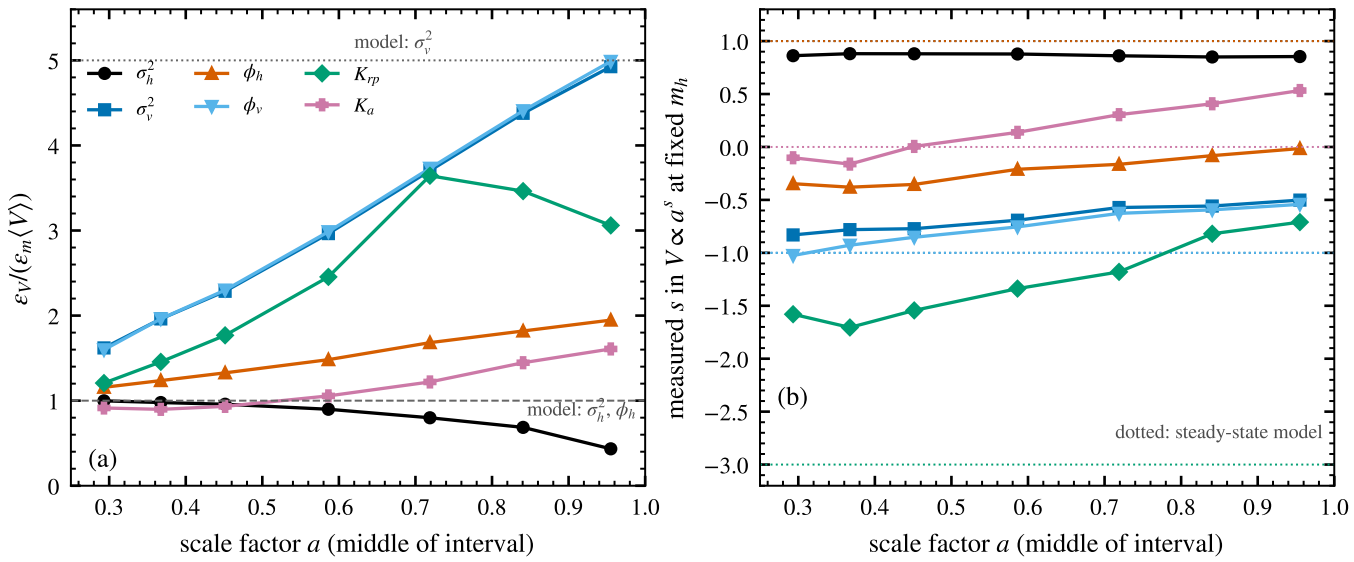


Figure 6. Test of equation (5). (a) $\varepsilon_V/(\varepsilon_m\langle V\rangle)$ for the six specific energies, against the steady-state model, $\varepsilon_V = \varepsilon_m\langle V\rangle$ for σ_h^2 and ϕ_h (dashed) and $5\varepsilon_m\langle V\rangle$ for σ_v^2 (dotted). (b) Measured growth exponent s of V in haloes of fixed mass (symbols) and the values of the steady-state model (dotted lines in the same colours).

(i) The kinetic energy of the motion of haloes has $s = 0.85$–0.88, close to the model value $s = 1$. But σ_h^2 decreases with n_p at late times, $\lambda_V < s$, and $\varepsilon_{kh}/(\varepsilon_m\langle\sigma_h^2\rangle)$ falls from 1.00 to 0.43: the approximation $\varepsilon_{kh} = \varepsilon_m\langle\sigma_h^2\rangle$ holds only for $z \geqslant 1$.

(ii) The inter-halo potential does not grow in haloes of fixed mass ($s = -0.35$ to -0.01; model: 1), and its mean grows only as $a^{0.34}$; $\varepsilon_{\phi h}/(\varepsilon_m\langle\phi_h\rangle)$ rises from 1.16 to 1.95.

(iii) For the internal energies, $s = -0.83$ to -0.50 (model: -1) and $\lambda_V \approx 1.4$ (model: 1). The two departures compensate: the ratio $\varepsilon_{kv}/(\varepsilon_m\langle\sigma_v^2\rangle)$ reaches 4.9 at $z \approx 0$, close to the model value 5.

(iv) The kinetic energy of radial motion decays in haloes of fixed mass much more slowly than the a^{-3} of the model ($s = -1.71$ to -0.71).

The fluxes are not constant in time: ε_{kh} falls from -0.37 to -0.05 and ε_{kv} levels off at about -0.6 after $z \approx 1$ (Table 2). A time-independent energy flux requires $\mu + \lambda_V - s$ constant and $\langle V\rangle M_h H$ constant; with the measured exponents the second condition is met only by the internal energies at $z < 0.5$.

6 RADIAL AND ROTATIONAL MOTION INSIDE HALOES

Inside a halo, part of the internal kinetic energy is coherent: radial infall and rotation. With x' and u' the physical position and peculiar velocity of a particle relative to its halo, the radial momentum and the angular momentum per unit mass are

$$G_{hp} = \frac{1}{n_p}\sum x'\cdot u', \qquad H_h = \frac{1}{n_p}\sum x'\times u', \tag{7}$$

and with the rms radius r_g we write $G_{hp} = -f_G a^{-1} H r_g^2$ and $|H_h| = f_H a^{1/2} H r_g^2$. The kinetic energies of radial and rotational motion are $K_{rp} = \frac{1}{2}(G_{hp}/r_g)^2$ and $K_a = \frac{1}{2}|H_h|\omega_h$, with the angular velocity $\omega_h = |H_h|/r_{rg}^2$ and the radius of gyration r_{rg} about the spin axis ($r_{rg}^2 = \frac{2}{3}r_g^2$ for a sphere). The rms radius grows as $r_g \approx c\,a\,(m_h/100m_p)^{1/3}\,\mathrm{Mpc}/h$ with $c = 0.30$ at $z = 0$, close to $c = 0.28$ for haloes of mean density $18\pi^2$ times the background and $r_g = r_h/\sqrt{3}$; at fixed mass $r_g \propto a^{0.88}$.

Rotation dominates small haloes. For pairs of particles $f_H \approx 1$1 and $f_G \approx 1.1$ at $z = 0$ (Fig. 7a); the ratio $\gamma_G = -G_{hp}a^{3/2}/|H_h|$ of pairs is 0.068–0.099 between $z = 2$ and 0, close to the value $1/(3\pi) = 0.106$ of the two-body collapse model [48]. For haloes of 100–1000 particles f_G and f_H are comparable ($\gamma_G = 0.83$ at $z = 2$, 1.07 at $z = 0$), and both drift with time: at fixed n_p, $-G_{hp}/(Hr_g^2) \propto a^{-0.6\text{ to }-0.8}$ and $|H_h|/(Hr_g^2) \propto a^{0.6\text{–}0.7}$. Large haloes rotate at $\omega_h/H = 1.2$, 1.7 and 2.5 at $z = 2$, 1 and 0 (Fig. 7b).

Their energies cascade inversely, but carry little. K_a changes little with time in haloes of fixed mass, while K_{rp} decreases (Fig. 8a); both grow with mass for $n_p \geqslant 20$. Both fluxes are negative (Fig. 8b). For haloes of 100–1000 particles K_a is 1.2 per cent of $|\phi_v|$ at $z = 0$, and $|\varepsilon_{ua}/\varepsilon_{\phi v}| = 0.8$ per cent; the radial flux is 2.4 times the rotational one. Coherent motion inside haloes thus takes a minor part in the energy cascade.

7 HALO SHAPE AND SPIN ORIENTATION

Measuring the shape. For each halo we compute the second-moment tensor of the particle positions about the centre of mass,

$$S_{ij} = \frac{1}{n_p}\sum_{p=1}^{n_p} x'_{p,i}x'_{p,j}. \tag{8}$$

The square roots of its eigenvalues, $r_{\lambda1} \leqslant r_{\lambda2} \leqslant r_{\lambda3}$, are the rms extents of the halo along its principal axes; $r_g^2 = r_{\lambda1}^2 + r_{\lambda2}^2 + r_{\lambda3}^2$, and for a uniform ellipsoid the semi-axes are $\sqrt{5}\,r_{\lambda i}$. From them we form the ellipticity, prolateness and triaxiality [19, 49, 50],

$$h_e = \frac{r_{\lambda3} - r_{\lambda1}}{2\Sigma}, \quad h_p = \frac{r_{\lambda3} - 2r_{\lambda2} + r_{\lambda1}}{2\Sigma}, \quad h_t = \frac{r_{\lambda3}^2 - r_{\lambda2}^2}{r_{\lambda3}^2 - r_{\lambda1}^2}, \tag{9}$$

with $\Sigma = r_{\lambda1} + r_{\lambda2} + r_{\lambda3}$: $h_p = h_e$ for a prolate and $h_p = -h_e$ for an oblate spheroid. The moment of inertia about the shortest axis,

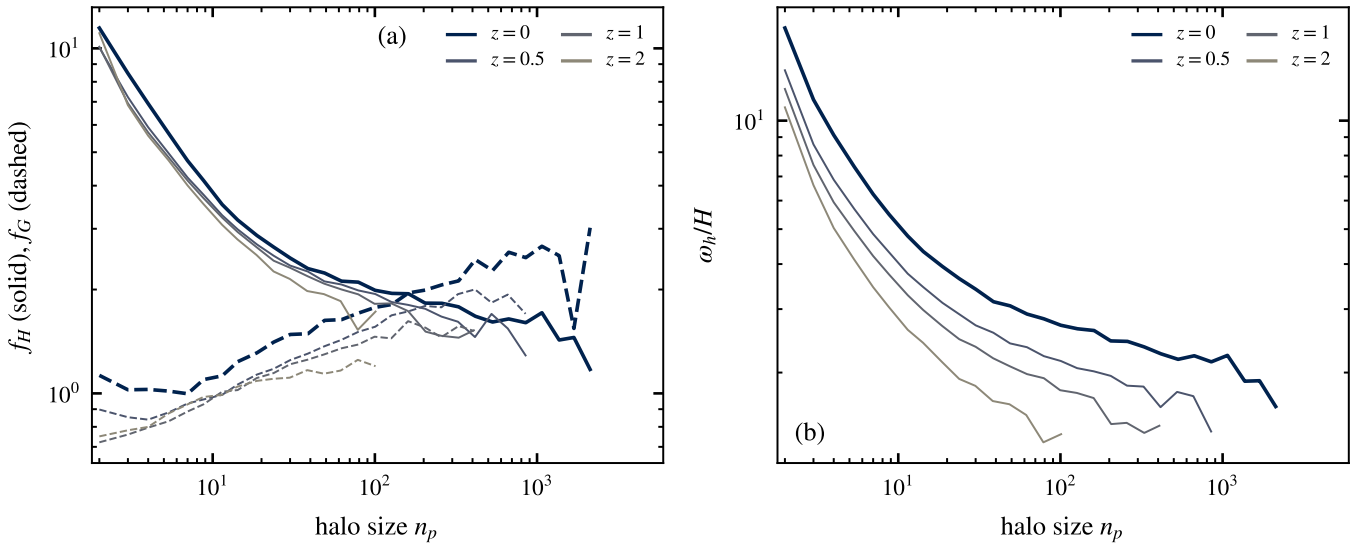


Figure 7. (a) Dimensionless radial momentum $f_G = -G_{hp}a/(Hr_g^2)$ (dashed) and angular momentum $f_H = |H_h|/(a^{1/2}Hr_g^2)$ (solid) of halo groups; (b) effective angular velocity ω_h in units of H. Small haloes rotate ($f_H \gg f_G$), large haloes fall in and rotate in comparable measure; large haloes spin at $\omega_h \approx 2$–$2.5H$ only at $z \leqslant 0.5$.

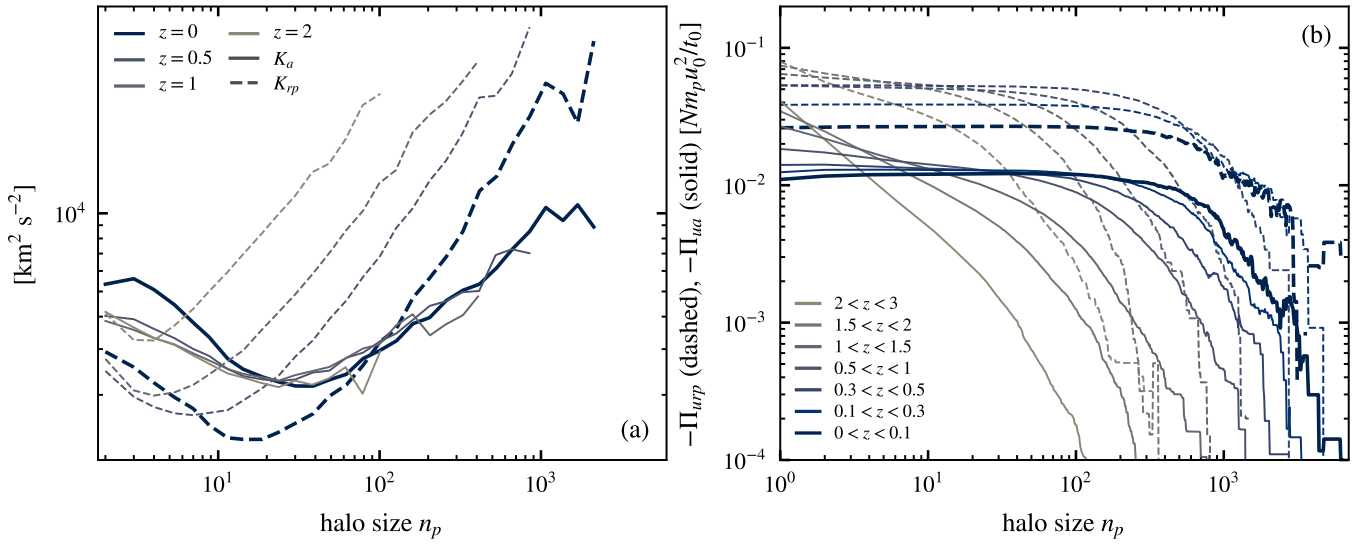


Figure 8. (a) Kinetic energy of the radial (K_{rp}, dashed) and rotational (K_a, solid) motion inside haloes; (b) their fluxes $-\Pi_{urp}$ (dashed) and $-\Pi_{ua}$ (solid). Both are carried to large haloes; the fluxes are about one per cent of the flux of the intra-halo potential.

relative to that of a sphere of the same volume, is

$$\lambda_{r2} = \frac{r_{\lambda3}^2 + r_{\lambda2}^2}{2\,(r_{\lambda1}r_{\lambda2}r_{\lambda3})^{2/3}}. \qquad (10)$$

Elongated, but with a nearly spherical moment of inertia. Haloes are elongated (Fig. 9): for $80 \leqslant n_p \leqslant 200$ the mean $r_{\lambda2}/r_{\lambda1}$ is 1.34, and $r_{\lambda3}/r_{\lambda1}$ falls to a minimum of 2.1 and rises to 2.7 for $2000 \leqslant n_p \leqslant 4000$. The mean triaxiality falls from 1 for pairs to 0.74 at $n_p \approx 24$ and returns to 0.83 for $n_p \geqslant 1000$: large haloes are mostly prolate. Yet the moment of inertia of single haloes with $n_p \geqslant 20$ exceeds that of a sphere of equal volume only by a median factor $\lambda_{r2} = 1.60$ (10–90 per cent: 1.30–2.23); 18 per cent of haloes exceed 2 and 1.8 per cent exceed 3. A stretched vortex, by contrast, can change its moment of inertia by large factors at fixed volume (by equation 10, a ratio $r_{\lambda3}/r_{\lambda1} = 10$ of a prolate body already gives $\lambda_{r2} \approx 11$).

A path through shape space. Two-particle haloes are lines, $h_e = h_p = \frac{1}{2}$; three-particle haloes are planar and lie exactly on $h_p = 3h_e - 1$ (mean $h_e = 0.373$, $h_p = 0.120$; Fig. 10). With growing size the group means move towards a sphere ($h_e = h_p = 0$) near the line $h_p = h_e/3$ (equivalently $r_{\lambda3} + 2r_{\lambda1} = 3r_{\lambda2}$, $\lambda_{r1} = \frac{1}{2}$), from which the means of $3 \leqslant n_p \leqslant 80$ deviate by at most 0.014. The smallest mean ellipticity, $\langle h_e \rangle = 0.119$ with $\langle h_p \rangle = 0.046$, is reached at $n_p \approx 111$; larger haloes are less spherical and follow a line parallel to the prolate limit, $\langle h_p \rangle = 1.00\langle h_e \rangle - 0.072$ ($200 \leqslant n_p \leqslant 2000$). Large

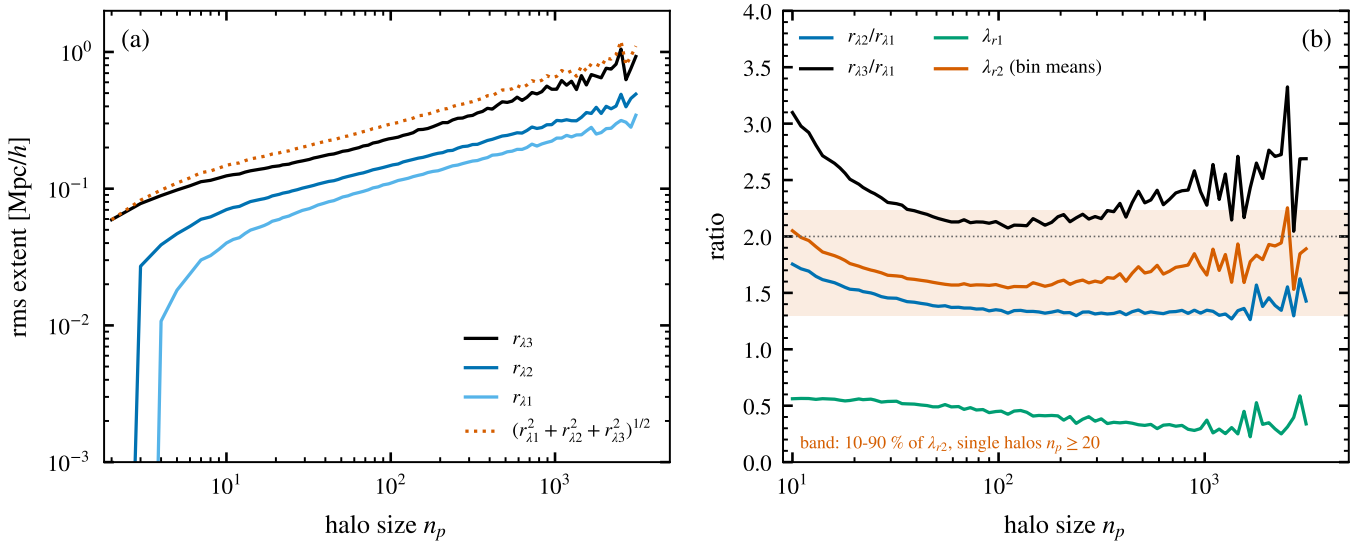


Figure 9. Principal extents of haloes at $z = 0$. (a) $r_{\lambda1}$–$r_{\lambda3}$ and r_g (bin means). (b) Axis ratios, $\lambda_{r1} = (r_{\lambda2} - r_{\lambda1})/(r_{\lambda3} - r_{\lambda2})$ and the moment-of-inertia ratio λ_{r2} of the bin means; the band is the 10–90 per cent range of λ_{r2} of single haloes with $n_p \geqslant 20$. Haloes are elongated, but their moment of inertia is close to that of a sphere.

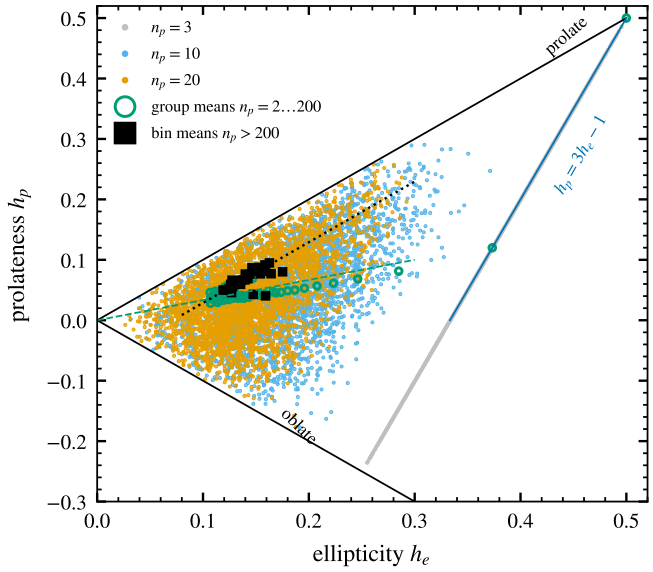


Figure 10. Haloes in the ellipticity–prolateness plane at $z = 0$: single haloes with $n_p = 3$, 10 and 20 (dots; 3000 of each), group means for $n_p = 2$ to 200 (circles) and bin means for $n_p > 200$ (squares). Black lines: prolate ($h_p = h_e$) and oblate ($h_p = -h_e$) limits; blue: three-particle haloes, $h_p = 3h_e - 1$; green dashed: $h_p = h_e/3$; dotted: fit $h_p = 1.00h_e - 0.072$ for $200 \leqslant n_p \leqslant 2000$.

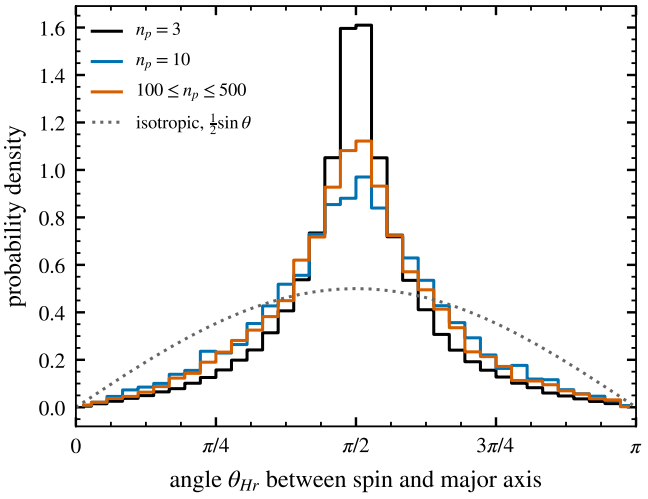


Figure 11. Probability density of the angle θ_{Hr} between the spin axis and the major axis at $z = 0$. Dotted: isotropic orientation. The spin axis of haloes is preferentially perpendicular to their major axis, most strongly for the smallest haloes.

haloes still accrete mass fast and have not relaxed; in time, haloes become rounder as accretion slows [17, 19].

Spin perpendicular to the major axis. The spin axis of a pair is exactly perpendicular to the line joining the particles, and this alignment is inherited as haloes grow by accreting single particles (Fig. 11): $|\theta_{Hr} - \pi/2| < \pi/6$ for 81 per cent of three-particle haloes, 71 per cent of ten-particle haloes and 70 per cent of haloes with $n_p \geqslant 20$ (isotropic: 50 per cent), in agreement with earlier simulations [14, 17, 25]. A stretched vortex instead spins about its long axis.

Conclusion on shape. A vortex passes energy to small scales because stretching at fixed volume and angular momentum lowers its moment of inertia. A halo grows in mass and volume, rotates about an axis perpendicular to its long axis, and its moment of inertia stays within a factor of two of that of a sphere. Its shape changes too little to drive an energy cascade; mass accretion carries the energy (Section 5).

8 DISCUSSION

What is firm. (1) The direction of the cascades: kinetic energy goes to large haloes and potential energy to small haloes, at all times and for all six energies in the propagation range (Fig. 5, Table 2). (2) Equation (5) with measured exponents (Fig. 6). (3) The shape and spin-orientation statistics (Section 7), which rest on 1.7 million haloes.

What is approximate. The propagation range is short (up to $n_p \sim$ 100–1000) and appears only at $z \leqslant 0.3$; the flux of σ_h^2 is not flat even then. The fluxes are finite differences between snapshots 0.06–0.19 t_0 apart.

The steady-state model. The steady state ($M_h \propto a^{1/2}$, energies $\propto a^{3/2}$, time-independent fluxes) is not reached by $z = 0$; the growth exponents decrease with time and may approach the steady values later. The model exponents for ϕ_h, σ_v^2 and K_{rp} disagree with the data, and $\varepsilon_V = \varepsilon_m \langle V \rangle$ holds for σ_h^2 and ϕ_h only at $z \geqslant 1$. The specific energy of the out-of-halo particles is small but changes with time and changes sign near $z \approx 1.7$.

Limitations. One simulation, an EdS SCDM cosmology with $m_p = 2.27 \times 10^{11} M_\odot/h$; the smallest haloes (2–20 particles) are below the resolution at which halo structure is reliable, and their statistics describe the particle dynamics of the simulation; FoF haloes include unbound particles; the potential is referenced to the $z = 10$ snapshot.

Tests. (1) Repeat the flux measurement in a ΛCDM simulation with finer snapshot spacing and higher resolution, and follow it into the future of an EdS run, to see whether $\mu \to \frac{1}{2}$ and $\lambda_V \to 1$. (2) Measure s directly by following haloes of fixed mass (merger trees). (3) Use spherical-overdensity haloes and unbinding to test the role of FoF bridges in the out-of-halo energy. (4) Compare the shape path of Fig. 10 with the evolution of individual haloes along their merger trees.

9 CONCLUSIONS

(i) The flux of any specific quantity through the halo mass scale is set by the mass flux and three growth exponents, $\varepsilon_V = -(\mu + \lambda_V - s) M_h H \langle V \rangle$; with the exponents measured in the simulation this holds to within 2 per cent for all six energies studied at $z < 2$.

(ii) Kinetic energy, of the motion of haloes and inside them, cascades to large haloes; inter- and intra-halo potential energy cascade to small haloes. Mass accretion of haloes carries both.

(iii) The simulation has not reached the steady state of the cascade picture: at $z \approx 0$ the halo mass grows as $a^{0.47}$ and the total energies as $a^{1.1\text{–}1.2}$, and the fluxes still change with time.

(iv) The coherent radial and rotational motion inside haloes also cascades to large haloes but carries about one per cent of the intra-halo potential flux.

(v) Haloes are elongated and spin about an axis perpendicular to their major axis, but their moment of inertia stays within a factor of about two of a sphere's; shape change is not the mechanism of the energy cascade.

DATA AVAILABILITY

The halo-group statistics are in the dark matter flow datasets on Zenodo [44, 45, 46], with presentation slides [51]. The scripts that compute every figure and number of this paper from these data, and the reduced per-halo shape catalogue at $z = 0$ are distributed with the paper.

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